MEDIUM · ADVANCED MATH · ANSWER KEY

Nonlinear equations in one variable and systems of equations in two variables

01

10 answers · Rationale-rich · Teach from doc

Q1 B

B. 48

Choice B is correct. Dividing each side of the given equation by 4 yields $\sqrt{2x} = 4$. Squaring both sides of this equation yields $2x = 16$. Multiplying each side of this equation by 3 yields $6x = 48$. Therefore, the value of $6x$ is 48.

Choice A is incorrect. This is the value of $3x$, not $6x$.

Choice C is incorrect. This is the value of $9x$, not $6x$.

Choice D is incorrect. This is the value of $12x$, not $6x$.

Q2 C

C. 8

Choice C is correct. The left-hand side of the given equation can be factored as $5x + 3x - 8$. Therefore, the given equation, $5x^2 - 37x - 24 = 0$, can be written as $5x + 3x - 8 = 0$. Applying the zero product property to this equation yields $5x + 3 = 0$ and $x - 8 = 0$. Subtracting 3 from both sides of the equation $5x + 3 = 0$ yields $5x = -3$. Dividing both sides of this equation by 5 yields $x = -\frac{3}{5}$. Adding 8 to both sides of the equation $x - 8 = 0$ yields $x = 8$. Therefore, the two solutions to the given equation, $5x^2 - 37x - 24 = 0$, are $-\frac{3}{5}$ and 8. It follows that 8 is the positive solution to the given equation.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q3

A

A. 0

Choice A is correct. Solutions to the given system of equations are ordered pairs (x,y) that satisfy both equations in the system. Adding the left-hand and right-hand sides of the equations in the system yields $x^2+y=6x+-6x+y+36$, or $x^2+y=y+36$. Subtracting y from both sides of this equation yields $x^2=36$. Taking the square root of both sides of this equation yields $x=6$ and $x=-6$. Therefore, there are two solutions to this system of equations, one with an x-coordinate of 6 and the other with an x-coordinate of -6. Substituting 6 for x in the second equation yields $y=-6(6)+36$, or $y=0$; therefore, one solution is $(6,0)$. Similarly, substituting -6 for x in the second equation yields $y=-6(-6)+36$, or $y=72$; therefore, the other solution is $(-6,72)$. It follows then that if (x,y) is a solution to the system, then possible values of xy are $(6)(0)=0$ and $(-6)(72)=-432$. Only 0 is among the given choices.

Choice B is incorrect. This is the x-coordinate of one of the solutions, $(6,0)$. Choice C is incorrect and may result from conceptual or computational errors. Choice D is incorrect. This is the square of the x-coordinate of one of the solutions, $(6,0)$.

Q4

C

C. 42

Choice C is correct. Multiplying both sides of the given equation by x yields $42 = 7x^2$. Therefore, the value of $7x^2$ is 42.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q5

A

A. $k = \frac{m - 14j}{5}$

Choice A is correct. Subtracting $14j$ from each side of the given equation results in $5k = m - 14j$. Dividing each side of this equation by 5 results in $k = \frac{m - 14j}{5}$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q6

B

B. $\frac{9}{4}$

Choice B is correct. Multiplying each side of the given equation by -16 yields $64x^2 + 112x = 576$. To complete the square, adding 49 to each side of this equation yields $64x^2 + 112x + 49 = 576 + 49$, or $8x + 7^2 = 625$. Taking the square root of each side of this equation yields two equations: $8x + 7 = 25$ and $8x + 7 = -25$. Subtracting 7 from each side of the equation $8x + 7 = 25$ yields $8x = 18$. Dividing each side of this equation by 8 yields $x = \frac{18}{8}$, or $x = \frac{9}{4}$. Therefore, $\frac{9}{4}$ is a solution to the given equation. Subtracting 7 from each side of the equation $8x + 7 = -25$ yields $8x = -32$. Dividing each side of this equation by 8 yields $x = -4$. Therefore, the given equation has two solutions, $\frac{9}{4}$ and -4. Since $\frac{9}{4}$ is positive, it follows that $\frac{9}{4}$ is the positive solution to the given equation.

Alternate approach: Adding $4x^2$ and $7x$ to each side of the given equation yields $0 = 4x^2 + 7x - 36$. The right-hand side of this equation can be rewritten as $4x^2 + 16x - 9x - 36$. Factoring out the common factor of $4x$ from the first two terms of this expression and the common factor of -9 from the second two terms yields $4xx + 4 - 9x + 4$. Factoring out the common factor of $x + 4$ from these two terms yields the expression $4x - 9x + 4$. Since this expression is equal to 0, it follows that either $4x - 9 = 0$ or $x + 4 = 0$. Adding 9 to each side of the equation $4x - 9 = 0$ yields $4x = 9$. Dividing each side of this equation by 4 yields $x = \frac{9}{4}$. Therefore, $\frac{9}{4}$ is a positive solution to the given equation. Subtracting 4 from each side of the equation $x + 4 = 0$ yields $x = -4$. Therefore, the given equation has two solutions, $\frac{9}{4}$ and -4. Since $\frac{9}{4}$ is positive, it follows that $\frac{9}{4}$ is the positive solution to the given equation.

Choice A is incorrect. Substituting $\frac{7}{4}$ for x in the given equation yields $-\frac{49}{2} = -36$, which is false.

Choice C is incorrect. Substituting 4 for x in the given equation yields $-92 = -36$, which is false.

Choice D is incorrect. Substituting 7 for x in the given equation yields $-245 = -36$, which is false.

Q7

D

D. $P = 100T + 40,000$

Choice D is correct. To express the value of the property in terms of the property tax, the given equation must be solved for P. Multiplying both sides of the equation by 100 gives $100T = P - 40,000$. Adding 40,000 to both sides of the equation gives $100T + 40,000 = P$. Therefore, $P = 100T + 40,000$.

Choice A is incorrect and may result from multiplying 40,000 by 0.01, then subtracting 400 from, instead of adding 400 to, the left-hand side of the equation. Choice B is incorrect and may result from multiplying 40,000 by 0.01. Choice C is incorrect and may result from subtracting instead of adding 40,000 from the left-hand side of the equation.

Q8

C

C. $-\frac{4}{5}$

Choice C is correct. Since a product of two factors is equal to 0 if and only if at least one of the factors is 0, either $5x + 4 = 0$ or $2x - 5 = 0$. Subtracting 4 from each side of the equation $5x + 4 = 0$ yields $5x = -4$. Dividing each side of this equation by 5 yields $x = -\frac{4}{5}$. Adding 5 to each side of the equation $2x - 5 = 0$ yields $2x = 5$. Dividing each side of this equation by 2 yields $x = \frac{5}{2}$. It follows that the solutions to the given equation are $-\frac{4}{5}$ and $\frac{5}{2}$. Therefore, $-\frac{4}{5}$ is a solution to the given equation.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q9

D

D. $n = \frac{16}{p - 20}$

Choice D is correct. To express n in terms of p, the given equation must be solved for n. Subtracting 9 from both sides of the given equation yields $p - 9 = \frac{14}{n}$. Since n is not equal to 0, multiplying both sides of this equation by n yields $p - 9n = 14$. It's given that $p > 9$, which means $p - 9$ is not equal to 0. Therefore, dividing both sides of $p - 9n = 14$ by $p - 9$ yields $\frac{p - 9n}{p - 9} = \frac{14}{p - 9}$, or $n = \frac{14}{p - 9}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q10

A

A. $(2,3)$
and $(-2,3)$

Choice A is correct. The two equations form a system of equations, and the solutions to the system correspond to the points of intersection of the graphs. The solutions to the system can be found by substitution. Since the second equation gives y = 3, substituting 3 for y in the first equation gives $3 = x^2 - 1$. Adding 1 to both sides of the equation gives $4 = x^2$. Solving by taking the square root of both sides of the equation gives $x = \pm 2$. Since y = 3 for all values of x for the second equation, the solutions are (2, 3) and (–2, 3). Therefore, the points of intersection of the two graphs are (2, 3) and (–2, 3).

Choices B, C, and D are incorrect and may be the result of calculation errors.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

Generated 2026-05-29 · cb-educator-question-bank